



Neuro-Symbolic RAG for Enterprise Applications: Bridging Inductive Tree-Based Reasoning with Deductive Graph-Structured Knowledge Representation

19CSE499 / 22AIE499 Project Phase 2 – Approval Presentation

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Problem Definition

What is the Problem?

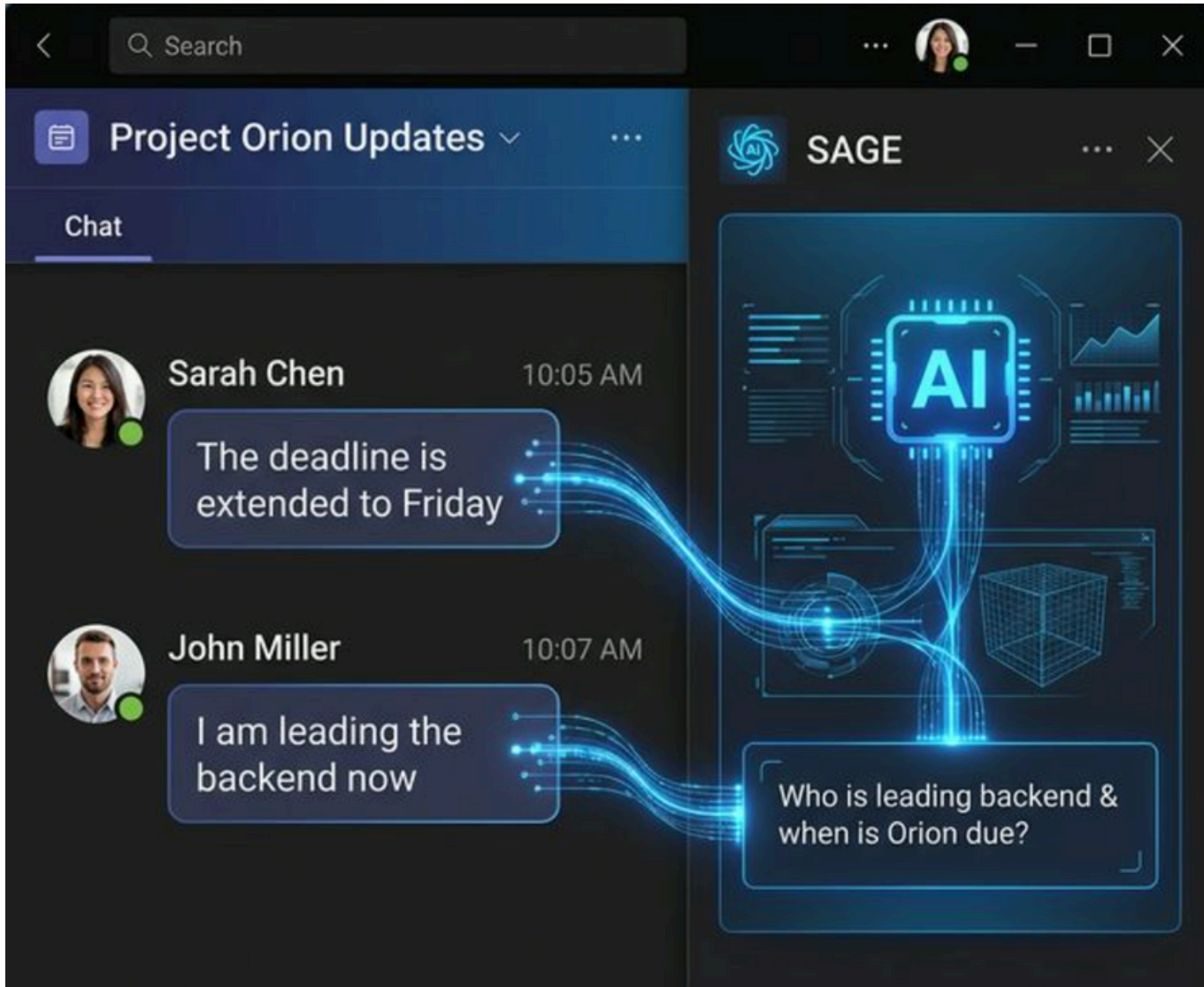
Systems return confident answers that are often ungrounded, unverifiable, and incompatible with enterprise rules.

- Ungrounded outputs / hallucinations: Answers may cite non-existent facts, docs, or relationships.
- Document-centric retrieval: Treats files as isolated, misses cross-document dependencies and multi-hop evidence.
- Single-pass RAG weakness: Complex, multi-step questions break down; errors compound.
- Lack of provenance & audit: Missing document IDs, node/edge references, and execution traces → not acceptable for compliance.

Why is this a Problem?

Unverifiable or incorrect answers cause legal, financial, and operational harm.

- Compliance & legal risk: Wrong or unsupported claims can breach regulations and incur fines.
- Loss of trust & ROI: Teams spend time validating outputs; automation value decreases.
- Operational inefficiency: Missed dependencies and duplicated work slow projects and reduce innovation.
- Security & privacy exposure: Unchecked answers risk leaking PII or violating policies.
- Measurable remediation targets: Grounding rate, graph path accuracy, hallucination rate, rule-satisfaction %, latency per stage.





The screenshot shows a Microsoft Teams chat interface. On the left is a sidebar with navigation icons for Activity, Chat, General, Design, Development, and Project Orion Updates (selected). The main chat area displays a conversation titled 'Project Orion Updates' with participants Sarah Chen, David Kim, John Miller, Emily Wong, and Sarah Chen. The chat history includes system notifications and messages about a deadline extension to Friday, a client meeting move, API doc updates, and a staging environment deployment check. A SAGE AI Assistant window is overlaid on the right, displaying a question and an answer based on the chat context.

Who is leading backend & when is Orion due?

Based on recent updates:

John Miller is now leading the backend team. The Project Orion deadline has been extended to this coming Friday.



Introduction

- Enterprises struggle to extract trusted insights from unstructured knowledge
PDFs, emails, chats, and documents remain siloed, unstructured, and difficult to reason over using traditional RAG systems.
- Traditional RAG systems retrieve text, but cannot reason over relationships
They lack multi-hop reasoning, organizational context, and verification—leading to hallucinations and shallow answers.
- SAGE introduces a graph-centric, enterprise-grade RAG architecture
It transforms unstructured enterprise data into a structured Neo4j knowledge graph and enables relationship-aware question answering.
- Combines Neural and Symbolic Intelligence
SAGE integrates LLMs and embeddings (neural) with knowledge graphs and constraints (symbolic) for more accurate and explainable responses.
- Designed for enterprise scale, auditability, and extensibility
Supports multi-model evaluation, performance benchmarking, provenance tracking, and future agentic reasoning.



Introduction

- SAGE (Self-Adaptive Graph RAG for Enterprises) is a graph-based Retrieval-Augmented Generation system that converts unstructured enterprise data into a knowledge graph and performs multi-hop, relationship-aware reasoning.
- It addresses key limitations of traditional RAG—hallucination, lack of structure, and poor explainability—by grounding LLM outputs in validated graph paths.
- Built using Neo4j, LangChain, FastAPI, and Groq-hosted LLMs, SAGE provides a scalable, extensible foundation for future agentic and neuro-symbolic AI systems.

Literature Survey

S.No.	Authors name(s)	Full title of the paper with year	Inference from the paper	Open Problem (For proposed work)
1.	Yazeed Alkhurayyif, Abdul Rahaman Wahab Sait	<i>Developing an Open Domain Arabic Question Answering System Using a Deep Learning Technique</i> (2023)	This paper presents a deep learning-based open-domain Arabic QA system using linguistic preprocessing and entity relationship extraction. The model achieves high accuracy on benchmark datasets, showing improved answer retrieval in low-resource language settings.	It lacks structured relationship modeling and explainable multi-hop reasoning required for enterprise-grade reliability.
2.	Zekai Shao, Shuran Sun, Yuheng Zhao, Siyuan Wang, Zhongyu Wei, Tao Gui, Cagatay Turkay, Siming Chen	Visual Explanation for Open-Domain Question Answering With BERT (2024)	The paper introduces VEQA, a visual analytics framework to explain BERT-based open-domain QA decisions. It helps experts understand retriever-reader behavior through attribution and visualization techniques.	The explanations are post-hoc and do not provide deterministic, auditable reasoning paths.

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3.	Chyi-Jiunn Lin, Guan-Ting Lin, Yung-Sung Chuang, Wei-Lun Wu, Shang-Wen Li, Abdelrahman Mohamed, Hung-yi Lee, Lin-shan Lee	SPEECHDPR: End-to-End Spoken Passage Retrieval for Open-Domain Spoken Question Answering (2024)	This work proposes SpeechDPR, an end-to-end dense retriever for open-domain spoken QA without relying on speech-to-text systems. It improves retrieval robustness, especially under poor ASR conditions.	The approach focuses on retrieval accuracy but does not address reasoning correctness or explainability.
4.	Chang Shu, Zijian Zhang, Youxin Chen, Jing Xiao, Jey Han Lau, Qian Zhang, Zheng Lu	Open Domain Response Generation Guided by Retrieved Conversations (2023)	The paper presents a hybrid retrieval-generation model that improves relevance in open-domain dialogue using keyword-guided transformers. Results show better response quality and diversity than existing methods.	The system prioritizes fluency over factual correctness, making it prone to hallucinations.

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5.	Maneeha Rani, Bhupesh Kumar Mishra, Dhavalkumar Thakker, Mohammad Nouman Khan	To Enhance Graph-Based Retrieval-Augmented Generation (RAG) with Robust Retrieval Techniques (2024)	This paper proposes a graph-based RAG framework combining keyword, vector, and graph retrieval to improve factual accuracy in healthcare LLMs. The approach reduces hallucination and improves source traceability.	The solution is domain-specific and does not generalize to complex enterprise-wide reasoning scenarios.
6.	Jie-Si Yang, Zhuoqi Zeng, Zijian Shen	Neural-Symbolic Dual-Indexing Architectures for Scalable Retrieval-Augmented Generation (2025)	This paper proposes a neural-symbolic dual-indexing RAG architecture that combines sparse knowledge graphs with keyword and vector retrieval to support scalable multi-hop reasoning. The approach reduces indexing cost while improving retrieval coverage and generation quality at enterprise scale.	Despite improved scalability, the framework does not fully address deterministic reasoning guarantees required for legal-grade enterprise decision systems.

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7.	Jiang Lai, Zheng Conghui, Zhang Xiaohan, Sun Fuhui, Wang Xiaoyan, Pan Li	Leveraging LLM based Retrieval-Augmented Generation for Legal Knowledge Graph Completion (2024)	This work introduces RA-KG-LLM, a retrieval-augmented generation approach for completing missing triples in legal knowledge graphs. By combining vector retrieval with LLM generation, the model improves relation prediction accuracy in judicial datasets.	The approach focuses on graph completion and does not provide end-to-end explainable reasoning paths for downstream enterprise question answering.
8.	Zhui Zhu, Guangpeng Qi, Guangyong Shang, Qingfeng He, Weichen Zhang, Ningbo Li, Yunzhi Chen, Lijun Hu, Wenqiang Zhang, Fan Dang	Enhancing Large Language Models with Knowledge Graphs for Robust Question Answering (2024)	This paper proposes SynaptiQA, a system that integrates LLMs with knowledge graphs to reduce hallucinations in domain-specific question answering. The model uses LLM-generated and optimized graph queries to improve accuracy and reliability.	The system depends on accurate query generation and does not guarantee consistent reasoning across complex multi-hop enterprise queries.

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9.	Feibo Jiang, Yubo Peng, Li Dong, Kezhi Wang, Kun Yang, Cunhua Pan, Dusit Niyato, Octavia A. Dobre	Large Language Model Enhanced Multi-Agent Systems for 6G Communications (2024)	This paper proposes CommLLM, a multi-agent LLM framework that integrates retrieval, planning, evaluation, and reflection to solve complex 6G communication tasks. The system improves adaptability and decision-making using collaborative agents.	The framework lacks structured knowledge grounding and explicit reasoning traceability for high-stakes decision validation.
10.	Kemou Jiang, Xuan Cai, Zhiyong Cui, Aoyong Li, Yilong Ren, Haiyang Yu, Hao Yang, Daocheng Fu, Licheng Wen, Pinlong Cai	KoMA: Knowledge-Driven Multi-Agent Framework for Autonomous Driving With Large Language Models (2025)	This paper presents KoMA, a knowledge-driven multi-agent framework using LLMs for autonomous driving, integrating multi-step planning, shared memory, and reflection. The system improves robustness and generalization in complex driving scenarios.	The framework is domain-specific and does not generalize to enterprise knowledge systems requiring deterministic reasoning and auditability.

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11.	Yinghao Sang	AutoCrit: A Meta-Reasoning Framework for Self-Critique and Iterative Error Correction in LLM Chains-of-Thought (2025)	AutoCrit is a meta-reasoning system that integrates reasoning, critique, and monitoring agents into a continuous feedback loop, enabling real-time error detection and correction and significantly reducing error propagation compared to post-hoc validation.	The framework relies heavily on the reliability of the Critique Agent, which can introduce spurious corrections if biased or hallucinating, leading to potential semantic drift
12.	Aakansha Mishra, Miriyala Srinivas Soumitri, Vikram N Rajendiran	Learning Representations from Explainable and Connectionist Approaches for Visual Question Answering (2024)	The authors propose a question-guided attention mechanism that fuses explainable symbolic reasoning (scene graphs) with connectionist multi-head attention transformer networks for VQA. The system learns to weight the importance of each modality based on the question type, improving accuracy and interpretability.	While effective for VQA, the model's performance on complex, long-form factual questions is limited by its reliance on internal world knowledge verification rather than external retrieval systems

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13.	Jeshmol P.J, Binsu C. Kovoov	A CLIP-based Video Question Answering framework with Explainable AI (2025)	This study introduces a VideoQA framework utilizing YOLOv8 for object detection and CLIP for semantic matching between queries and video frames. It incorporates the LIME algorithm to enhance the interpretability of similarity-based frame selection.	The system currently focuses on short video clips (1-20 seconds) and lacks the ability to generate descriptive, long-form reasoning trajectories for open-ended queries
14.	Sunny Dhokane, Siddhesh Karande, Chinmay Deshmukh, Bhakti Karangale, Akshit Bollabattin, Pradip S. Varade	BM25 Implementation For Information Retrieval: Candidate Shortlister For Recruitment Process (2024)	This paper implements the Best Match 25 (BM25) ranking algorithm for candidate shortlisting, integrating custom parsers and query processors to align job requirements with resumes. Empirical results demonstrate that BM25 outperforms traditional TF-IDF by accounting for document length normalization and term saturation.	The system lacks a natural language generation component to explain why a candidate was ranked highly, making the final decision less transparent for HR professionals



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15	Liu Xingguang, Cheng Zhenbo, Shen Zhengyuan, Zhang Haoxin, Meng Hangcheng, Xu Xuesong, Xiao Gang	Building a Question Answering System for the Manufacturing Domain (2022)	The study presents a domain-specific QA system using template-based question generation and an IA-BiLSTM model to enhance sentence similarity matching, enabling fast and accurate access to pressure vessel design standards with 98.15% accuracy.	The system lacks flexibility, as it cannot answer queries that do not closely match predefined templates, limiting reasoning over unstructured and unseen documents.
16	Yumeng Sun, Lian Lian, Renhan Zhang, Heyi Wang, Renzi Meng, Xuehui Quan	Fusion-Based Retrieval-Augmented Generation for Complex Question Answering with LLMs (2025)	The paper proposes a fusion-based RAG framework that combines structured knowledge graphs and unstructured text through dual-channel retrieval, improving factual correctness and reasoning depth in complex QA tasks.	Linguistic fluency and syntactic consistency remain challenging, as reflected by relatively low BLEU scores across specialized domains.



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17	Yunshi Lan, Gaole He, Jinhao Jiang, Jing Jiang, Wayne Xin Zhao, Ji-Rong Wen	Complex Knowledge Base Question Answering: A Survey (2023)	This survey systematically reviews complex KBQA techniques, classifying them into semantic parsing-based and information retrieval-based methods, and highlights key challenges such as multi-hop reasoning, large search spaces, and learning under weak supervision.	Existing KBQA systems lack post-deployment learning from failure cases and exhibit limited compositional generalization when encountering unseen schemas or novel query structures.
18	Binita Saha, Utsha Saha, Muhammad Zubair Malik	QuIM-RAG: Advancing Retrieval-Augmented Generation With Inverted Question Matching for Enhanced QA Performance (2024)	The paper proposes QuIM-RAG, which generates candidate questions from document chunks and applies inverted question matching to improve semantic retrieval accuracy, thereby reducing hallucinations and context dilution in RAG systems.	The approach does not yet support real-time or automatic updates, limiting its effectiveness for frequently changing or dynamic web-based knowledge sources.



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19	Rohan Gamidi, Vinitha Chowdary A, Tejaswi Muppala, Priyanka C. Nair	SAGE: Self-Adaptive Graph for Enterprise Retrieval-Augmented Generation (2025)	The paper proposes SAGE, a GraphRAG-based framework that converts unstructured enterprise data into a dynamic Enterprise Knowledge Graph (EKG) using Neo4j. It employs hybrid retrieval combining vector similarity search with multi-hop graph traversal to provide richer contextual grounding for LLMs such as LLaMA-3.3.	The system lacks an automated mechanism for dynamic graph schema evolution, limiting adaptability as enterprise structures and relationships change over time.
20	Tienlan Sun, Anaiy Somalwar, Hinson Chan	Multimodal Retrieval Augmented Generation Evaluation Benchmark (2024)	This work introduces a multimodal evaluation benchmark for RAG systems, incorporating MRAFE to extract and embed features from both text and images using YOLO-v8, enabling robust testing against noise, negative rejection, and counterfactual scenarios.	The benchmark is evaluated primarily on a proprietary education-focused dataset (CRISDE600), and its generalizability to other specialized or industry-agnostic domains remains insufficiently validated.



Addressing Research Gaps

6

- Most existing RAG systems rely only on semantic similarity search

Literature primarily focuses on vector-based retrieval, which fails to capture explicit relationships and dependencies present in enterprise data.

- Limited use of knowledge graphs for reasoning, not retrieval

Prior works using knowledge graphs mainly treat them as auxiliary retrieval sources, not as core reasoning structures.

- Inadequate support for multi-hop reasoning

Surveyed approaches struggle with queries requiring multiple inference steps across entities, leading to incomplete or shallow answers.

- High hallucination risk due to lack of verification mechanisms

Existing RAG models generate responses without validating whether the reasoning path actually exists in the data

- Weak integration of symbolic constraints with neural models

Most literature focuses on neural LLM-based reasoning, with minimal incorporation of symbolic rules, constraints, or schemas.

- Limited explainability and provenance tracking

Prior systems rarely provide clear evidence paths, making answers difficult to audit or trust in enterprise environments.

- Lack of systematic performance comparison frameworks

- Many studies propose architectures but do not rigorously compare graph-based RAG against traditional RAG using standardized metrics.



How SAGE Addresses these gaps

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- Uses a Neo4j knowledge graph as a first-class reasoning layer, not just a retrieval index
- Enables multi-hop, relationship-aware reasoning across documents, messages, and entities
- Grounds LLM responses in validated graph paths, reducing hallucinations
- Combines neural embeddings with symbolic graph constraints (neuro-symbolic approach)
- Provides structured provenance and evaluation metrics for enterprise and research validation
-



Justification for the Proposed Problem

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1. Knowledge Graph–Based Enterprise Modeling

- Problem: Enterprise data has complex relationships that flat document retrieval cannot capture.
- Our Approach: We model enterprise entities and their relationships in a knowledge graph for structured understanding.

2. Multi-Hop Reasoning Through Agentic Planning

- Problem: Business queries often require multiple reasoning steps, which traditional RAG cannot perform.
- Our Approach: An agentic planner decomposes complex questions into smaller, logical steps.

3. Graph-Constrained Reasoning

- Problem: LLMs may assume incorrect links between entities, leading to hallucinations.
- Our Approach: Reasoning paths are restricted to valid relationships present in the knowledge graph.

4. Hybrid Retrieval (Graph + Semantic Search)

- Problem: Using only vector search misses structured connections, while only graphs miss document context.
- Our Approach: We combine graph traversal with semantic document retrieval for more complete answers.



Justification for the Proposed Problem

5. Hallucination Reduction via Data Grounding

- Problem: AI systems may generate confident but unsupported information.
- Our Approach: Answers are generated only after verifying evidence from trusted enterprise data sources.

6. Built-in Policy and Rule Enforcement

- Problem: Traditional systems may give answers that violate company policies or regulations.
- Our Approach: Organizational rules are checked and enforced before answers are returned.

7. Explainable Answers with Provenance

- Problem: Enterprises cannot trust answers that lack transparency.
- Our Approach: Every response includes traceable sources, graph paths, and supporting data.

8. Agent-Based Verification (Critic Loop)

- Problem: Incorrect or weakly supported answers may still pass through the system.
- Our Approach: A critic agent reviews outputs for grounding, correctness, and policy compliance before final delivery.

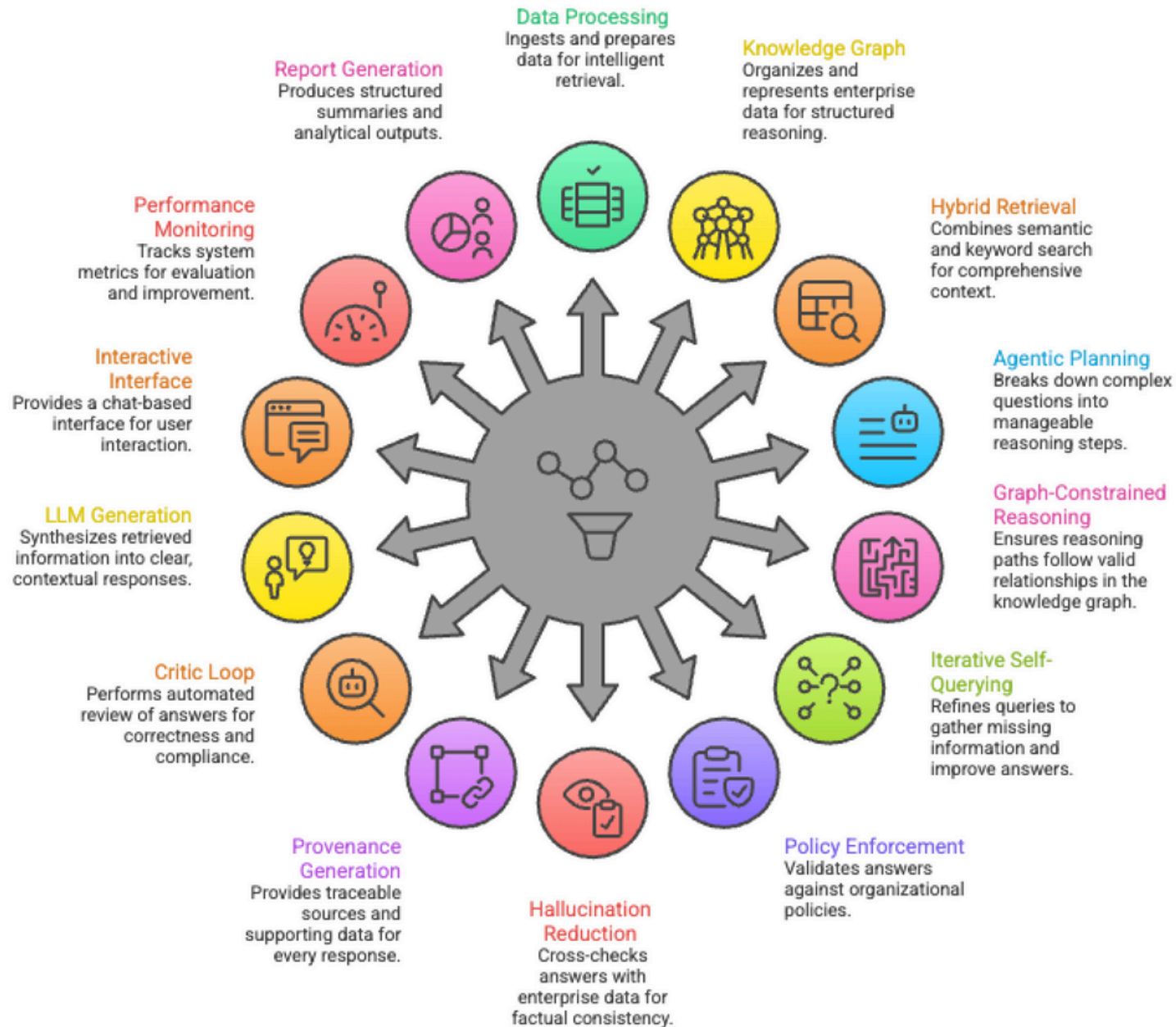


Architecture of the Proposed System

- Agentic orchestrator (brain): Planner → Executor → Critic; coordinates tool calls and loops until verified.
- Hybrid retrieval layer: Dense vectors + BM25 + Graph traversal; reranker + de-duplication.
- Symbolic layer: Neo4j path validator, policy_guard rule engine, and schema constraints (G-CT / GAP).
- Generator & Critic: Produce answer + provenance; critic enforces grounding and may trigger replan.
- Integration points: agentic_mode toggle in backend.py/graph_rag.py, audit logs, and provenance bundle (plan JSON, node IDs, Cypher paths).

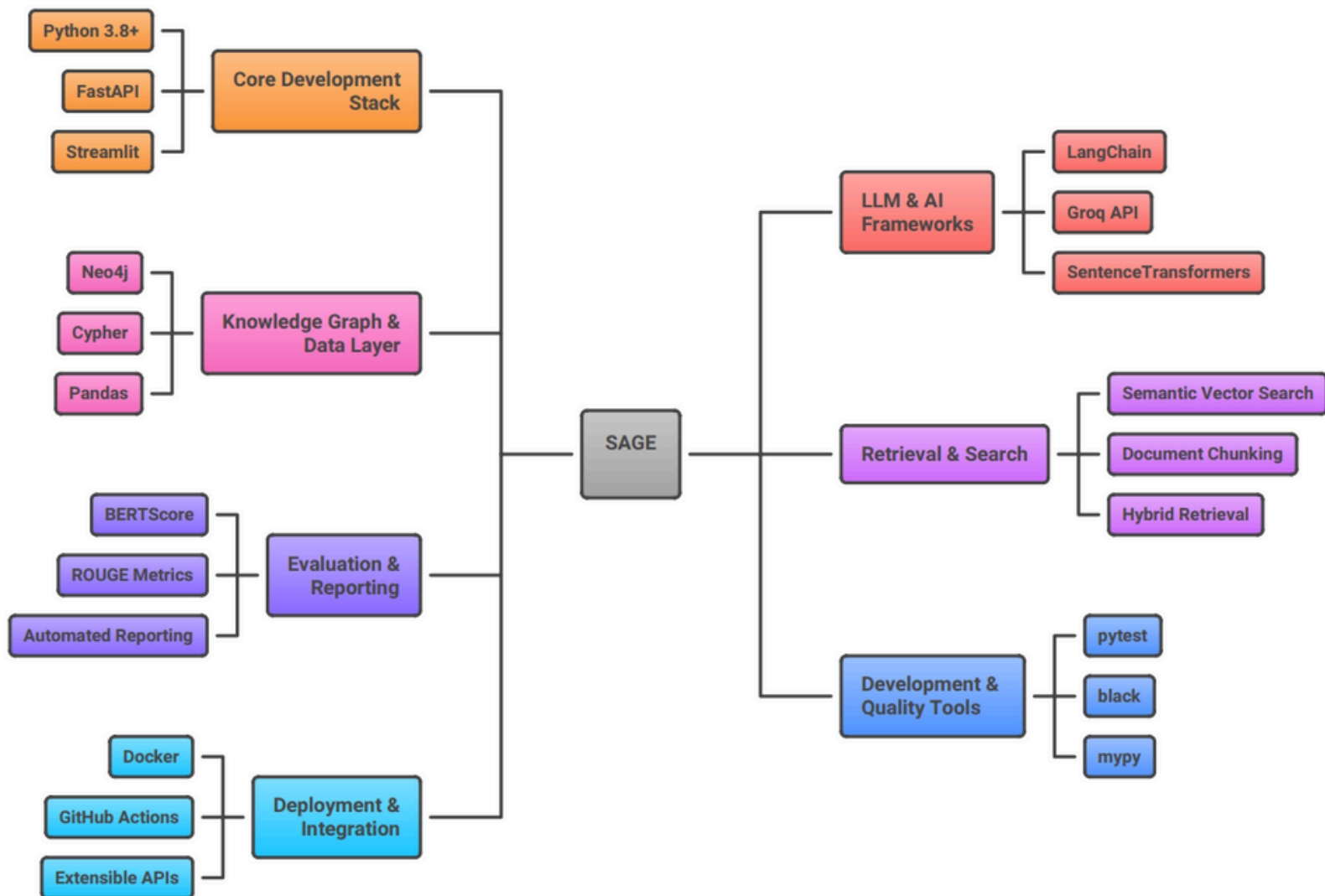


Functionalities of the Proposed System



Software/Tools Requirements

13



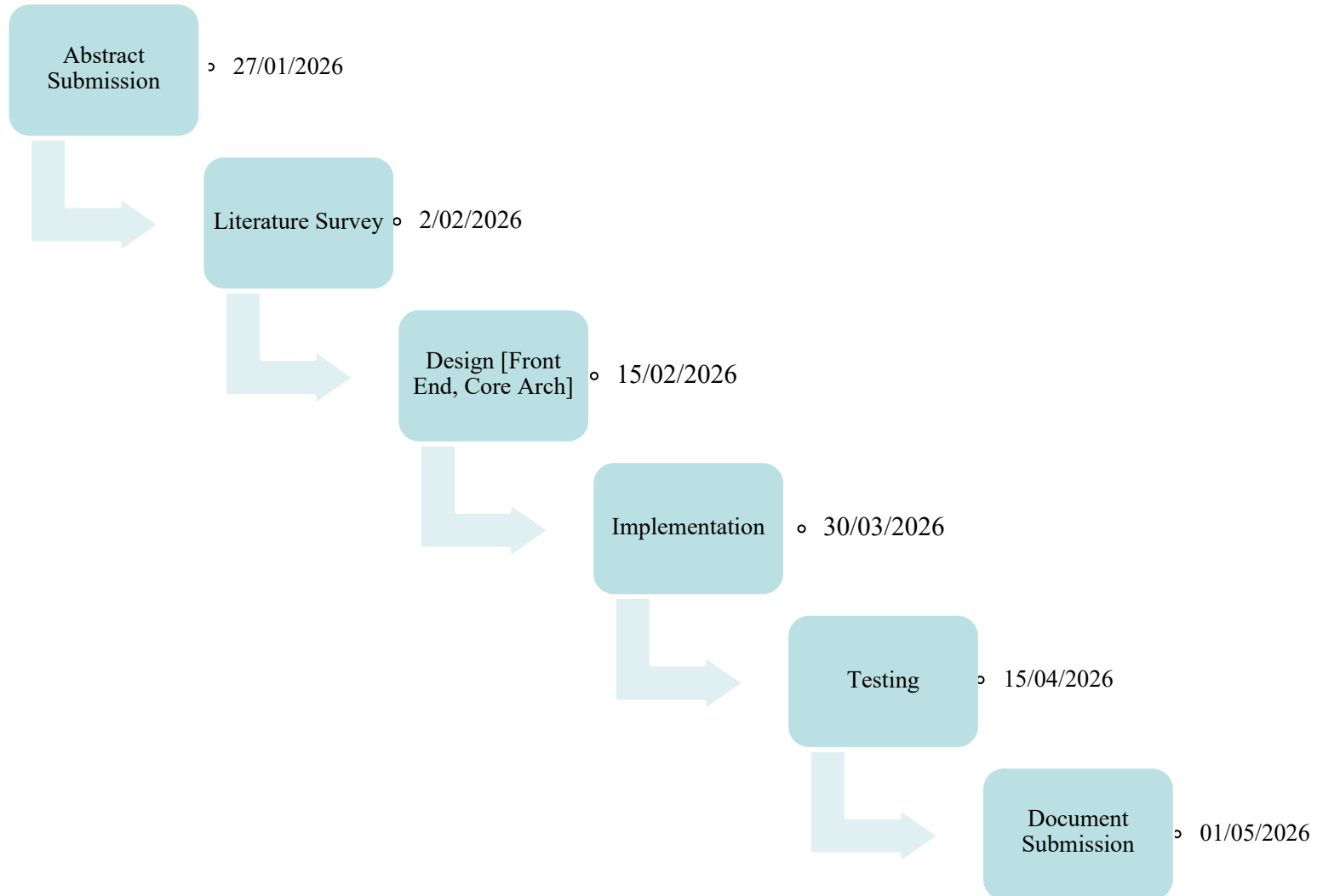


Novelty

14

- SAGE introduces a graph-first Retrieval-Augmented Generation architecture where the knowledge graph acts as the core reasoning layer rather than auxiliary storage.
- SAGE unifies documents, messages, entities, and relationships into a single Neo4j knowledge graph to enable holistic enterprise-level reasoning across heterogeneous data.
- SAGE enables explicit multi-hop, relationship-aware reasoning over unstructured data, which is not supported by traditional vector-based RAG systems.
- SAGE implements a neuro-symbolic approach by combining LLM-based embeddings with symbolic graph structures and constraints to improve reliability and reduce hallucinations.
- SAGE performs graph-grounded answer generation using validated graph paths and structured contextual evidence to ensure answers are based on real enterprise knowledge.
- SAGE includes a built-in empirical evaluation framework that quantitatively compares graph-based RAG with traditional RAG across quality, latency, and similarity metrics.
- SAGE provides an extensible architectural foundation for agent-driven, explainable, and verifiable AI systems suitable for enterprise and research-grade applications.

Timeline



Conclusion

References

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THANK YOU